

MAINTENANCE GUIDE

FANUC ROBODRILL a-D21MiB5

Compact Machining Center - Maintenance

This is sample documentation generated for an Atlas CMMS demonstration environment. It mirrors the structure of a real OEM document but does not contain proprietary manufacturer content.

1. Safety Precautions

Read and understand all safety information before operating or servicing this machine.

- Always lock out / tag out (LOTO) the main disconnect before maintenance.
- Wait for all spindle and axis motion to stop before opening guards.
- Wear approved eye protection and safety footwear in the machining area.
- Keep the work envelope clear of tools, fixtures, and personnel during operation.
- Never defeat door interlocks or safety light curtains.

2. Routine Maintenance Schedule

Daily:

- Check coolant level and concentration; top off as required.
- Wipe down way covers and inspect for chips and leaks.
- Verify air pressure at the FRL and drain the air filter bowl.

Weekly:

- Clean the chip conveyor and coolant tank strainer.
- Inspect way wipers and telescoping covers for damage.

Monthly:

- Check and top off the way lube reservoir (use OEM-specified oil only).
- Inspect the spindle drawbar and ATC operation.

Annually:

- Replace way lube and hydraulic oil; service the spindle as required.

3. Lubrication Specifications

Way lube: ISO VG 68 way oil (e.g. Mobil Vactra No. 2).

Hydraulic: ISO VG 32 (e.g. Mobil DTE 25).

Spindle oil: ISO VG 10 spindle oil (e.g. Mobil Velocite No. 10).

- Follow the lubrication chart and intervals in the maintenance schedule.

4. Troubleshooting

Alarm: low lubrication pressure -> check reservoir level and pump operation.

Symptom: poor surface finish -> check spindle runout, tool holders, and way lube.

Symptom: axis following error -> inspect ball screw, bearings, and servo drive.

Symptom: coolant flow reduced -> inspect pump impeller and filter screens.